

Credit Name: CSE2920 - CSE Project C

Assignment Name: Read Button

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to modify the given code to print the state of the green button, only when pressed. I will also observe how the program reacts when the loop repeats every 10ms, 1000ms, and 5000ms. Explanations of how I modified the code are below:

CODING:

1.

```
//Add Phidgets Library | You added a file called phidget22 when c
import com.phidget22.*;

public class ReadButton {
    //Handle Exceptions | Exceptions will happen in your code fro
    public static void main(String[] args) throws Exception{

        //Create | Here you've created a DigitalInput object for
        DigitalInput greenButton = new DigitalInput();

        //Address | This tells your program where to find the dev
        greenButton.setHubPort(5);
        greenButton.setIsHubPortDevice(true);
    }
}
```

Phidgets library is imported

GreenButton object was created to interact with the physical phidgets box.

setHubPort was changed to 5, because I need to modify the code to read the GreenButton instead of the Red Button. The GreenButton hub number is 5.

IsHubPortDevice set to true (confirming to the program that the GreenButton is part of phidgets box).

2.

```
//Open | Open establishes a connection between your object and your
greenButton.open(1000);

//Use your Phidgets | Here is where you use your Phidgets! This code
while(true){
    if(greenButton.getState()) {
        System.out.println("Button State: " + greenButton.getState());
        Thread.sleep(1000);
    }
}
```

Opened the GreenButton, giving the program 1 second to locate the phidget (established a connection to the physical Phidgets box).

While statement used to loop the code.

If the green button is pressed, then the state true is returned. Print statement only works if there is a button press.

Loop time at 10ms makes the button press very sensitive, because it is checking so often.

Loop at 1000ms misses some of my button presses, because it only checks for button presses every second.

Loop at 5000ms misses most of my button presses, because it only checks for button presses every 5 seconds.

End of Program!